Grade 11 – Term 1
C9: Normal versus Student's t, with Errors

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1 Models and calculation tools

Open the normal and Student's t reference tables. Page 1 gives $\Phi(z) = P(Z \leq z)$ for positive z on a hundredth grid (through 2.59). Page 2 gives positive t critical values for upper-tail probabilities $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$, for df 1–30 and selected larger df. Those entries do not give arbitrary CDF values. Use a suitable calculator's t CDF and inverse t CDF for this activity; normal CDF and inverse normal are also allowed. Numerical CDF and quantile results in the solutions are supplementary to the tables. Never turn table bounds into a single probability or error.

Write $F_\nu(t) = P(T \leq t)$ for $T \sim t_\nu$. Symmetry gives $F_\nu(-c) = 1 - F_\nu(c)$; a right tail is $1 - F_\nu(c)$. An interval has probability $F_\nu(b) - F_\nu(a)$ and its outside complement has probability $F_\nu(a) + 1 - F_\nu(b)$. Replace F_ν by Φ for normal calculations. Endpoints have zero probability. For central area p, each tail has $(1-p)/2$; for combined symmetric tails q, each tail has $q/2$.

A contextual location-scale model is explicitly $X = m + aT$, with $a > 0$ a scale parameter; a is not automatically the standard deviation. Thus $T = (X - m)/a$. These are stipulated models for signed deviations, not claims that arbitrary individual measurements follow t. For independent random observations from a normal population, with unknown population standard deviation, the sampling statistic instead is

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Here $\bar{X}$ and S vary across samples; do not treat observed s as a fixed population standard deviation.

Attempt each task before checking the worked solution. Calculate with full precision, then report probabilities to four decimals, standardized quantiles to three decimals, contextual values to two decimals, and percentage errors to two decimals. Displayed rounded intermediate values are not the inputs used for errors. Diagrams shade the requested region; curves continue beyond the displayed range.

C9. Compare normal approximations with t probabilities and quantiles, and interpret their errors. For each comparison define $Z \sim N(0,1)$ and $T \sim t_\nu$. The normal calculation approximates the t calculation. In contextual units use

$X_N = m + aZ$ and $X_T = m + aT$, with exactly the same m and a. Probability comparisons keep the same boundaries and event; inverse comparisons keep the same target probability. With normal result r_N and nonzero t reference r_T, calculate

$$A = |r_N - r_T|, \qquad R = \frac{|r_N - r_T|}{|r_T|}, \qquad E = 100R\%.$$

For intervals compare each endpoint separately. These errors depend on the quantity compared: a standardized quantile and a contextual cutoff need not have the same relative error. Heavier t tails give larger probabilities beyond positive cutoffs and more extreme quantiles at the same tail probability. As df increases, t approaches normal. An interval entirely on one side of zero need not contain less t probability.

2 Direct probability problems

Problems:

2.1. Delivery schedule deviations

2.2. Production cost deviations

2.3. Electricity forecast deviations

2.4. Monthly sales deviations

2.5. Commute forecast deviations

2.6. Revenue planning alerts

2.1 Delivery schedule deviations

Problem. A planning team models a delivery time minus its scheduled time, measured in minutes. Use $Z \sim N(0,1)$ and $T \sim t_9$, with $X_N = 0 + 5Z$ and $X_T = 0 + 5T$. The scale is 5 minutes.

Questions/Tasks

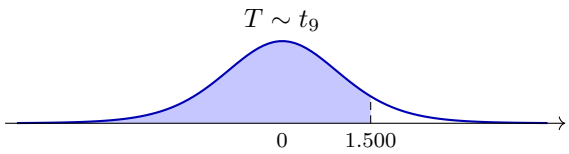
C9 Find $P(X < 7.5)$, and $P(X < -6)$. Calculate each probability under both models, its absolute error, relative error and percentage relative error, and interpret the normal approximation.

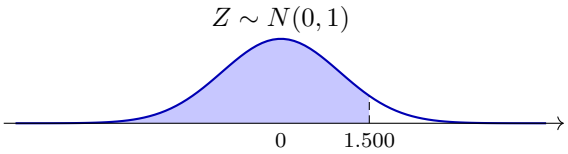
Worked solution

Standardize each boundary using $(x-0)/5$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(X_T < 7.5) = P(T < 1.5).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). The region is cumulative area to the left. Supplementary CDF values: $F_9(1.5) \approx 0.9160747$.

$$P(T < 1.5) = F_9(1.5) \approx 0.9161.$$

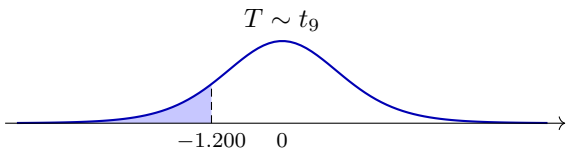
$$P(Z < 1.5) = \Phi(1.5) \approx 0.9332.$$

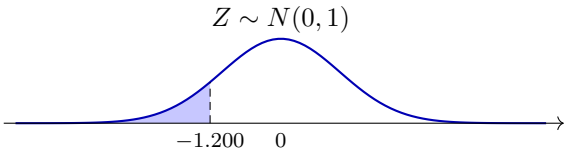
$$A = |0.9331928 - (0.9160747)| \approx 0.017118, \quad R = \frac{A}{0.9160747} \approx 0.018686, \quad E \approx 1.87\%.$$

Interpretation. About 91.61% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 1.71 percentage points (1.87% relative to t).

Part 2.

$$P(X_T < -6) = P(T < -1.2).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). The region is cumulative area to the left. By symmetry, $F_9(-1.2) = 1 - F_9(1.2) \approx 1 - 0.8696134 = 0.1303866$. Supplementary CDF values: $F_9(-1.2) \approx 0.1303866$.

$$P(T < -1.2) = F_9(-1.2) \approx 0.1304.$$

$$P(Z < -1.2) = \Phi(-1.2) \approx 0.1151.$$

$$A = |0.1150697 - (0.1303866)| \approx 0.015317, \quad R = \frac{A}{0.1303866} \approx 0.117473, \quad E \approx 11.75\%.$$

Interpretation. About 13.04% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 1.53 percentage points (11.75% relative to t).

2.2 Production cost deviations

Problem. A planning team models a batch cost minus its budget, measured in monetary units. Use $Z \sim N(0,1)$ and $T \sim t_{14}$, with $X_N = 0 + 20Z$ and $X_T = 0 + 20T$. The scale is 20 monetary units.

Questions/Tasks

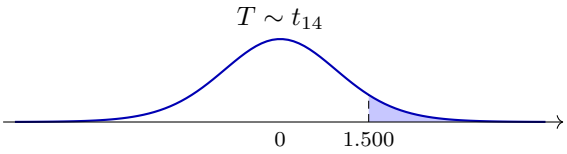
C9 Find $P(X > 30)$, and $P(X > -16)$. Calculate each probability under both models, its absolute error, relative error and percentage relative error, and interpret the normal approximation.

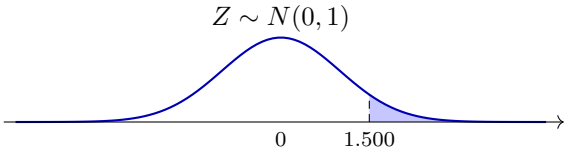
Worked solution

Standardize each boundary using $(x - 0)/20$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(X_T > 30) = P(T > 1.5).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Take the complement of the cumulative area. Supplementary CDF values: $F_{14}(1.5) \approx 0.9220873$.

$$P(T > 1.5) = 1 - F_{14}(1.5) \approx 0.0779.$$

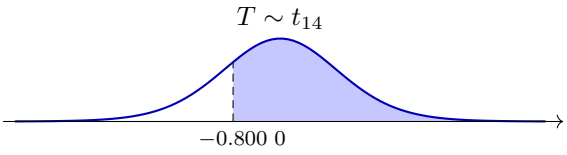
$$P(Z > 1.5) = 1 - \Phi(1.5) \approx 0.0668.$$

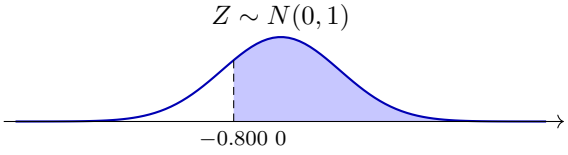
$$A = |0.0668072 - (0.0779127)| \approx 0.011105, \quad R = \frac{A}{0.0779127} \approx 0.142537, \quad E \approx 14.25\%.$$

Interpretation. About 7.79% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 1.11 percentage points (14.25% relative to t).

Part 2.

$$P(X_T > -16) = P(T > -0.8).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Take the complement of the cumulative area. By symmetry, $F_{14}(-0.8) = 1 - F_{14}(0.8) \approx 1 - 0.7814622 = 0.2185378$. Supplementary CDF values: $F_{14}(-0.8) \approx 0.2185378$.

$$P(T > -0.8) = 1 - F_{14}(-0.8) \approx 0.7815.$$

$$P(Z > -0.8) = 1 - \Phi(-0.8) \approx 0.7881.$$

$$A = |0.7881446 - (0.7814622)| \approx 0.006682, \quad R = \frac{A}{0.7814622} \approx 0.008551, \quad E \approx 0.86\%.$$

Interpretation. About 78.15% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 0.67 percentage points (0.86% relative to t).

2.3 Electricity forecast deviations

Problem. A planning team models household use minus its forecast, measured in kWh. Use $Z \sim N(0,1)$ and $T \sim t_{19}$, with $X_N = 0 + 50Z$ and $X_T = 0 + 50T$. The scale is 50 kWh.

Questions/Tasks

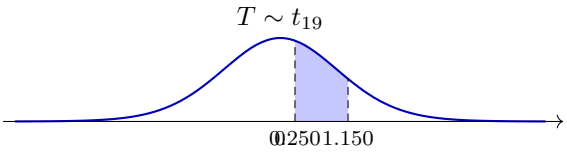
C9 Find $P(12.5 < X < 57.5)$. Calculate each probability under both models, its absolute error, relative error and percentage relative error, and interpret the normal approximation.

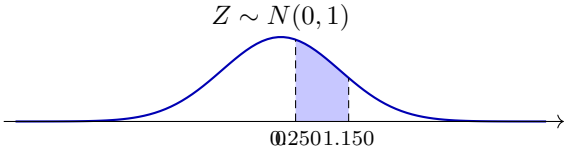
Worked solution

Standardize each boundary using $(x - 0)/50$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(12.5 < X_T < 57.5) = P(0.25 < T < 1.15).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Subtract the left cumulative area from the right cumulative area. Supplementary CDF values: $F_{19}(0.25) \approx 0.5973648$, $F_{19}(1.15) \approx 0.8677933$.

$$P(0.25 < T < 1.15) = F_{19}(1.15) - F_{19}(0.25) \approx 0.2704.$$

$$P(0.25 < Z < 1.15) = \Phi(1.15) - \Phi(0.25) \approx 0.2762.$$

$$A = |0.2762217 - (0.2704285)| \approx 0.005793, \quad R = \frac{A}{0.2704285} \approx 0.021422, \quad E \approx 2.14\%.$$

Interpretation. About 27.04% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 0.58 percentage points (2.14% relative to t).

2.4 Monthly sales deviations

Problem. A planning team models monthly sales minus their target, measured in units. Use $Z \sim N(0,1)$ and $T \sim t_{11}$, with $X_N = 0 + 20Z$ and $X_T = 0 + 20T$. The scale is 20 units.

Questions/Tasks

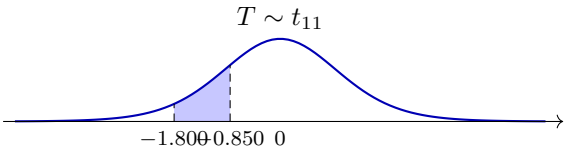
C9 Find $P(-36 < X < -17)$. Calculate each probability under both models, its absolute error, relative error and percentage relative error, and interpret the normal approximation.

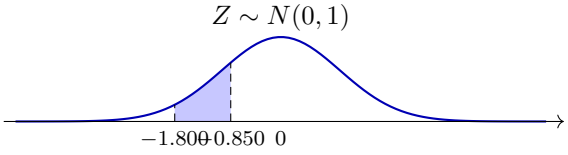
Worked solution

Standardize each boundary using $(x - 0)/20$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(-36 < X_T < -17) = P(-1.8 < T < -0.85).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{11}(-1.8) = 1 - F_{11}(1.8) \approx 1 - 0.9503421 = 0.0496579$. By symmetry, $F_{11}(-0.85) = 1 - F_{11}(0.85) \approx 1 - 0.7932783 = 0.2067217$. Supplementary CDF values: $F_{11}(-1.8) \approx 0.0496579$, $F_{11}(-0.85) \approx 0.2067217$.

$$P(-1.8 < T < -0.85) = F_{11}(-0.85) - F_{11}(-1.8) \approx 0.1571.$$

$$P(-1.8 < Z < -0.85) = \Phi(-0.85) - \Phi(-1.8) \approx 0.1617.$$

$$A = |0.1617322 - (0.1570638)| \approx 0.004668, \quad R = \frac{A}{0.1570638} \approx 0.029723, \quad E \approx 2.97\%.$$

Interpretation. About 15.71% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 0.47 percentage points (2.97% relative to t).

2.5 Commute forecast deviations

Problem. A planning team models a commute time minus its forecast, measured in minutes. Use $Z \sim N(0,1)$ and $T \sim t_{24}$, with $X_N = 0 + 10Z$ and $X_T = 0 + 10T$. The scale is 10 minutes.

Questions/Tasks

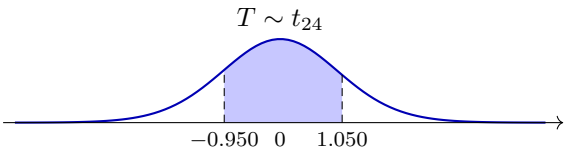
C9 Find $P(-9.5 < X < 10.5)$. Calculate each probability under both models, its absolute error, relative error and percentage relative error, and interpret the normal approximation.

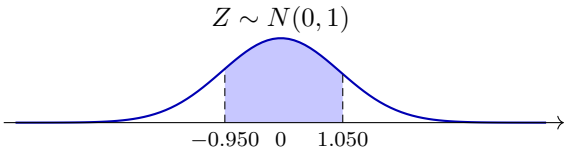
Worked solution

Standardize each boundary using $(x - 0)/10$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(-9.5 < X_T < 10.5) = P(-0.95 < T < 1.05).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{24}(-0.95) = 1 - F_{24}(0.95) \approx 1 - 0.8242092 = 0.1757908$. Supplementary CDF values: $F_{24}(-0.95) \approx 0.1757908$, $\quad F_{24}(1.05) \approx 0.8479116$.

$$P(-0.95 < T < 1.05) = F_{24}(1.05) - F_{24}(-0.95) \approx 0.6721.$$

$$P(-0.95 < Z < 1.05) = \Phi(1.05) - \Phi(-0.95) \approx 0.6821.$$

$$A = |0.6820848 - (0.6721208)| \approx 0.009964, \quad R = \frac{A}{0.6721208} \approx 0.014825, \quad E \approx 1.48\%.$$

Interpretation. About 67.21% of model outcomes satisfy this planning condition under t. The normal approximation overestimates that probability, by 1.00 percentage points (1.48% relative to t).

2.6 Revenue planning alerts

Problem. A planning team models weekly revenue minus its plan, measured in thousand monetary units. Use $Z \sim N(0,1)$ and $T \sim t_5$, with $X_N = 0 + 8Z$ and $X_T = 0 + 8T$. The scale is 8 thousand monetary units.

Questions/Tasks

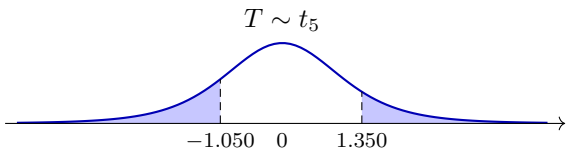
C9 Find $P(X < -8.4 \text{ or } X > 10.8)$, and $P(X < -16 \text{ or } X > 16)$. Calculate each probability under both models, its absolute error, relative error and percentage relative error, and interpret the normal approximation.

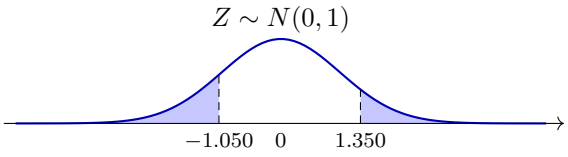
Worked solution

Standardize each boundary using $(x-0)/8$. In the task, X means X_T for the t model and X_N for the normal approximation.

Part 1.

$$P(X_T < -8.4 \text{ or } X_T > 10.8) = P(T < -1.05 \text{ or } T > 1.35).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Add the disjoint outside regions, or subtract the intervening interval from one. By symmetry, $F_5(-1.05) = 1 - F_5(1.05) \approx 1 - 0.8291023 = 0.1708977$. Supplementary CDF values: $F_5(-1.05) \approx 0.1708977$, $F_5(1.35) \approx 0.8825456$.

$$P(T < -1.05 \text{ or } T > 1.35) = F_5(-1.05) + 1 - F_5(1.35) \approx 0.2884.$$

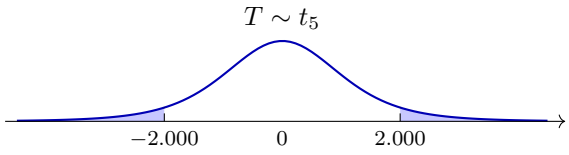
$$P(Z < -1.05 \text{ or } Z > 1.35) = \Phi(-1.05) + 1 - \Phi(1.35) \approx 0.2354.$$

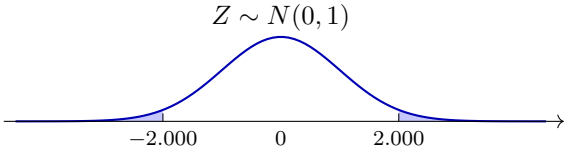
$$A = |0.2353670 - (0.2883521)| \approx 0.052985, \quad R = \frac{A}{0.2883521} \approx 0.183751, \quad E \approx 18.38\%.$$

Interpretation. About 28.84% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 5.30 percentage points (18.38% relative to t).

Part 2.

$$P(X_T < -16 \text{ or } X_T > 16) = P(T < -2 \text{ or } T > 2).$$





For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$ (and $\Phi(-c) = 1 - \Phi(c)$). Add the disjoint outside regions, or subtract the intervening interval from one. By symmetry, $F_5(-2) = 1 - F_5(2) \approx 1 - 0.9490303 = 0.0509697$. Supplementary CDF values: $F_5(-2) \approx 0.0509697$, $F_5(2) \approx 0.9490303$.

$$P(T < -2 \text{ or } T > 2) = F_5(-2) + 1 - F_5(2) \approx 0.1019.$$

$$P(Z < -2 \text{ or } Z > 2) = \Phi(-2) + 1 - \Phi(2) \approx 0.0455.$$

$$A = |0.0455003 - (0.1019395)| \approx 0.056439, \quad R = \frac{A}{0.1019395} \approx 0.553654, \quad E \approx 55.37\%.$$

Interpretation. About 10.19% of model outcomes satisfy this planning condition under t. The normal approximation underestimates that probability, by 5.64 percentage points (55.37% relative to t).

3 Inverse probability problems

Problems:

3.1. Top delivery delays

3.2. Bottom cost deviations

3.3. Central energy deviations

3.4. Central sales deviations

3.5. Two-tail checks on filling weights

3.6. Two-tail checks on processing times

3.1 Top delivery delays

Problem. Model delivery time minus its schedule in minutes by $X_T = 0 + 5T$, $T \sim t_9$. The normal approximation uses $X_N = 0 + 5Z$, $Z \sim N(0,1)$, with the same location and scale. Find the cutoff exceeded by the top 5%.

Questions/Tasks

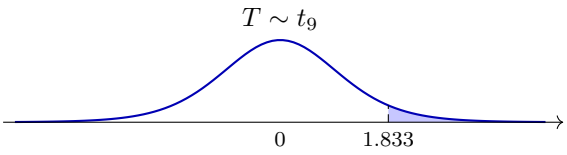
C9 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Under both models calculate the results, then absolute, relative and percentage relative errors for each standardized endpoint separately. Interpret the approximation.

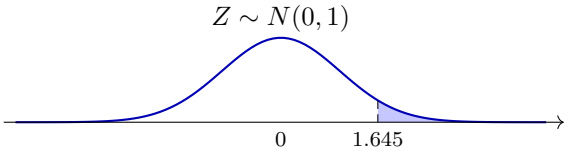
Worked solution

$$P(X_T > x) = 0.0500 \quad \Longrightarrow \quad P(T \le c) = 0.9500, \qquad c = \frac{x - 0}{5}.$$

A top percentage is a right tail, so subtract it from one before inverse CDF.

$$c_T = F_9^{-1}(0.95) \approx 1.833, \qquad c_N = \Phi^{-1}(0.95) \approx 1.645.$$





$$A = |1.6448536 - (1.8331129)| \approx 0.188259, \quad R = \frac{A}{1.8331129} \approx 0.102699, \quad E \approx 10.27\%.$$

$$x_T = 0 + 5c_T \approx 9.17 \text{ minutes}, \qquad x_N = 0 + 5c_N \approx 8.22 \text{ minutes}.$$

Interpretation. About 5% of deviations are above 9.17 minutes. Normal places the cutoff closer to the location, so it understates the magnitude of this extreme deviation. Errors above compare standardized quantiles.

3.2 Bottom cost deviations

Problem. Model batch cost minus its budget in monetary units by $X_T = 0 + 20T$, $T \sim t_{11}$. The normal approximation uses $X_N = 0 + 20Z$, $Z \sim N(0,1)$, with the same location and scale. Find the cutoff below which the bottom 10% fall.

Questions/Tasks

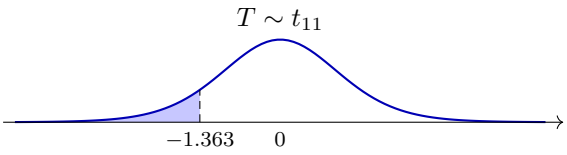
C9 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Under both models calculate the results, then absolute, relative and percentage relative errors for each standardized endpoint separately. Interpret the approximation.

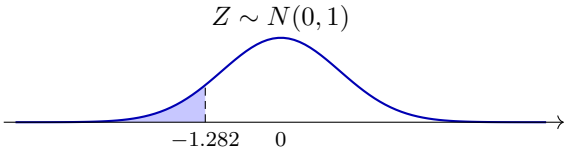
Worked solution

$$P(X_T < x) = 0.1000 \quad \Longrightarrow \quad P(T \le c) = 0.1000, \qquad c = \frac{x - 0}{20}.$$

The bottom percentage is already a left cumulative probability; the cutoff is negative by symmetry.

$$c_T = F_{11}^{-1}(0.1) \approx -1.363, \qquad c_N = \Phi^{-1}(0.1) \approx -1.282.$$





$$A = |-1.2815516 - (-1.3634303)| \approx 0.081879, \quad R = \frac{A}{1.3634303} \approx 0.060053, \quad E \approx 6.01\%.$$

$$x_T = 0 + 20c_T \approx -27.27 \text{ monetary units}, \qquad x_N = 0 + 20c_N \approx -25.63 \text{ monetary units}.$$

Interpretation. About 10% of deviations are below -27.27 monetary units. Normal places the cutoff closer to the location, so it understates the magnitude of this extreme deviation. Errors above compare standardized quantiles

3.3 Central energy deviations

Problem. Model electricity use minus its forecast in kWh by $X_T = 0 + 50T$, $T \sim t_{19}$. The normal approximation uses $X_N = 0 + 50Z$, $Z \sim N(0, 1)$, with the same location and scale. Find the symmetric interval about 0 containing the middle 95%.

Questions/Tasks

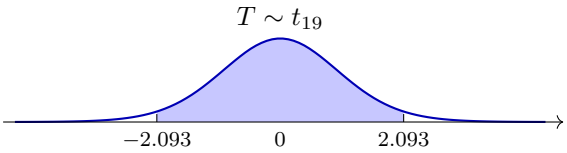
C9 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Under both models calculate the results, then absolute, relative and percentage relative errors for each standardized endpoint separately. Interpret the approximation.

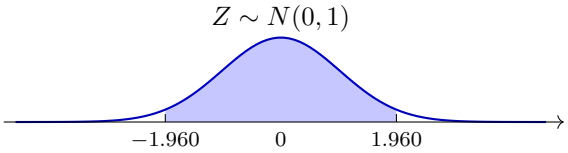
Worked solution

$$P(-c < T < c) = 0.9500 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0250.$$

Divide the outside complement $1 - p$ by two because symmetry makes the tails equal.

$$F_{19}(c) = 0.9750, \qquad c_T = F_{19}^{-1}(0.975) \approx 2.093, \quad c_N = \Phi^{-1}(0.975) \approx 1.960.$$





The standardized endpoints are $[-2.093, 2.093]$ under t and $[-1.960, 1.960]$ under normal. Compare the corresponding lower endpoints:

$$A = |-1.9599640 - (-2.0930241)| \approx 0.133060, \quad R = \frac{A}{2.0930241} \approx 0.063573, \quad E \approx 6.36\%.$$

Compare the corresponding upper endpoints:

$$A = |1.9599640 - (2.0930241)| \approx 0.133060, \quad R = \frac{A}{2.0930241} \approx 0.063573, \quad E \approx 6.36\%.$$

$$L_T = 0 - 50c_T \approx -104.65, \qquad U_T = 0 + 50c_T \approx 104.65 \quad \text{(kWh)}.$$

Normal gives $[-98.00, 98.00]$ kWh. Errors above compare standardized endpoints, not these contextual values. **Interpretation.** The t interval contains 95% of modeled deviations; 2.5% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.

3.4 Central sales deviations

Problem. Model sales minus their target in units by $X_T = 0 + 20T$, $T \sim t_{29}$. The normal approximation uses $X_N = 0 + 20Z$, $Z \sim N(0,1)$, with the same location and scale. Find the symmetric interval about 0 containing the middle 80%.

Questions/Tasks

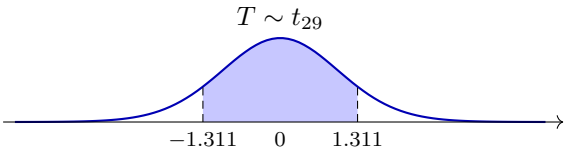
C9 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Under both models calculate the results, then absolute, relative and percentage relative errors for each standardized endpoint separately. Interpret the approximation.

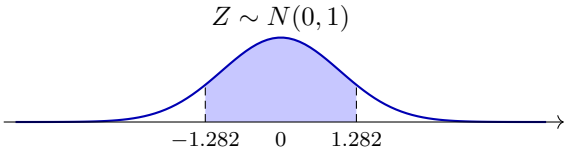
Worked solution

$$P(-c < T < c) = 0.8000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.1000.$$

Divide the outside complement $1 - p$ by two because symmetry makes the tails equal.

$$F_{29}(c) = 0.9000, \qquad c_T = F_{29}^{-1}(0.9) \approx 1.311, \quad c_N = \Phi^{-1}(0.9) \approx 1.282.$$





The standardized endpoints are $[-1.311, 1.311]$ under t and $[-1.282, 1.282]$ under normal. Compare the corresponding lower endpoints:

$$A = |-1.2815516 - (-1.3114336)| \approx 0.029882, \quad R = \frac{A}{1.3114336} \approx 0.022786, \quad E \approx 2.28\%.$$

Compare the corresponding upper endpoints:

$$A = |1.2815516 - (1.3114336)| \approx 0.029882, \quad R = \frac{A}{1.3114336} \approx 0.022786, \quad E \approx 2.28\%.$$

$$L_T = 0 - 20c_T \approx -26.23, \qquad U_T = 0 + 20c_T \approx 26.23 \quad \text{(units)}.$$

Normal gives $[-25.63, 25.63]$ units. Errors above compare standardized endpoints, not these contextual values. **Interpretation.** The t interval contains 80% of modeled deviations; 10% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.

3.5 Two-tail checks on filling weights

Problem. A quality team takes independent random samples of $n = 6$ filling weights from a normal population with mean μ and unknown standard deviation. Define $T = (\bar{X} - \mu)/(S/\sqrt{6}) \sim t_5$. Find symmetric limits for the standardized sample mean leaving a combined two-tail probability of 0.1000. Compare with $Z \sim N(0, 1)$ as an approximation to $T \sim t_5$ for the same target probability.

Questions/Tasks

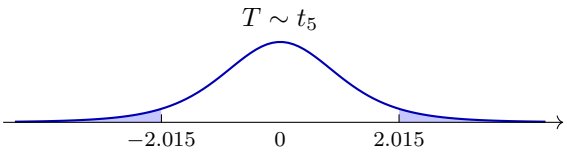
C9 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s). Under both models calculate the results, then absolute, relative and percentage relative errors for each standardized endpoint separately. Interpret the approximation.

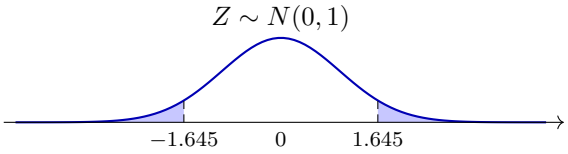
Worked solution

$$P(|T| > c) = 0.1000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0500.$$

Divide the combined outside area by two because symmetry makes the tails equal.

$$F_5(c) = 0.9500, \qquad c_T = F_5^{-1}(0.95) \approx 2.015, \quad c_N = \Phi^{-1}(0.95) \approx 1.645.$$





The standardized endpoints are $[-2.015, 2.015]$ under t and $[-1.645, 1.645]$ under normal. Compare the corresponding lower endpoints:

$$A = |-1.6448536 - (-2.0150484)| \approx 0.370195, \quad R = \frac{A}{2.0150484} \approx 0.183715, \quad E \approx 18.37\%.$$

Compare the corresponding upper endpoints:

$$A = |1.6448536 - (2.0150484)| \approx 0.370195, \quad R = \frac{A}{2.0150484} \approx 0.183715, \quad E \approx 18.37\%.$$

Interpretation. The t interval contains 90% of standardized sample-mean outcomes; 5% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.

3.6 Two-tail checks on processing times

Problem. A quality team takes independent random samples of $n = 25$ processing times from a normal population with mean μ and unknown standard deviation. Define $T = (\bar{X} - \mu)/(S/\sqrt{25}) \sim t_{24}$. Find symmetric limits for the standardized sample mean leaving a combined two-tail probability of 0.0400. Compare with $Z \sim N(0, 1)$ as an approximation to $T \sim t_{24}$ for the same target probability.

Questions/Tasks

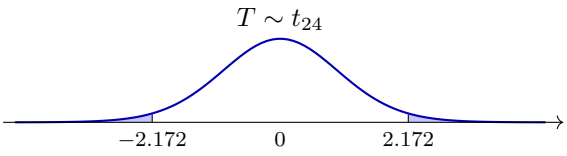
C9 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s). Under both models calculate the results, then absolute, relative and percentage relative errors for each standardized endpoint separately. Interpret the approximation.

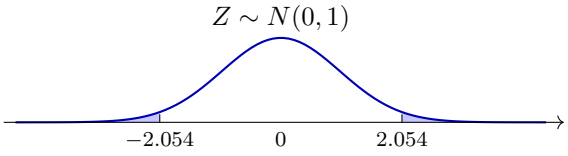
Worked solution

$$P(|T| > c) = 0.0400 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0200.$$

Divide the combined outside area by two because symmetry makes the tails equal.

$$F_{24}(c) = 0.9800, \qquad c_T = F_{24}^{-1}(0.98) \approx 2.172, \quad c_N = \Phi^{-1}(0.98) \approx 2.054.$$





The standardized endpoints are $[-2.172, 2.172]$ under t and $[-2.054, 2.054]$ under normal. Compare the corresponding lower endpoints:

$$A = |-2.0537489 - (-2.1715447)| \approx 0.117796, \quad R = \frac{A}{2.1715447} \approx 0.054245, \quad E \approx 5.42\%.$$

Compare the corresponding upper endpoints:

$$A = |2.0537489 - (2.1715447)| \approx 0.117796, \quad R = \frac{A}{2.1715447} \approx 0.054245, \quad E \approx 5.42\%.$$

Interpretation. The t interval contains 96% of standardized sample-mean outcomes; 2% lie in each excluded tail. The normal limits are less extreme because normal has lighter tails; using them for t would trigger more alerts or cover less than the intended central percentage.

4 How sample size improves the approximation

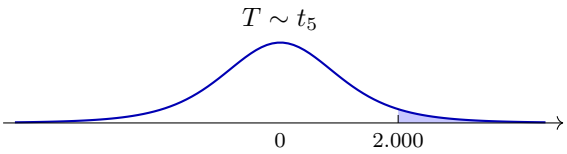
Problem. A filling-weight audit takes independent random samples from a normal population with unknown standard deviation. For each $n = 6, 20, 60, 120$, let $T = (\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$ and compare with $Z \sim N(0, 1)$.

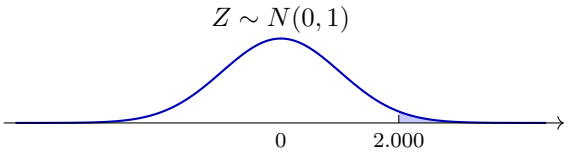
Questions/Tasks

C9 For every sample size, calculate the probability of exceeding 2 under both distributions and the upper cutoff exceeded in 2.5 percent of outcomes under both distributions. Calculate all three errors for each probability and each quantile. Interpret the trend for audit alerts.

Worked solution

Keep the event fixed: $P(T > 2) = 1 - F_\nu(2)$ and $P(Z > 2) = 1 - \Phi(2)$. Keep the inverse target fixed: $P(T > c_T) = P(Z > c_N) = 0.025$, so use cumulative area 0.975. Here $\nu = n - 1$; the rows 59 and 119 require calculator supplements.





Probability comparison. Normal gives $1 - \Phi(2) \approx 0.0228$.

n	ν	t result	normal result	A	R	E (%)
6	5	0.0510	0.0228	0.028220	0.553654	55.37
20	19	0.0300	0.0228	0.007251	0.241688	24.17
60	59	0.0251	0.0228	0.002305	0.092000	9.20
120	119	0.0239	0.0228	0.001139	0.047665	4.77

Quantile comparison. Normal gives $\Phi^{-1}(0.975) \approx 1.960$.

n	ν	t result	normal result	A	R	E (%)
6	5	2.571	1.960	0.610618	0.237541	23.75
20	19	2.093	1.960	0.133060	0.063573	6.36
60	59	2.001	1.960	0.041031	0.020505	2.05
120	119	1.980	1.960	0.020136	0.010169	1.02

For example, in either table $A = |r_N - r_T|$, $R = A/|r_T|$, and $E = 100R\%$, using unrounded calculator results. **Interpretation.** All three errors decrease across these sample sizes. Normal underestimates the chance of an audit statistic exceeding 2 and understates the cutoff required for a 2.5% alert rate. Heavier t tails explain both effects; increasing df reduces the difference. Estimating the population standard deviation still calls for t, even when the approximation is close.